Module 6: Answer Selection and the Full Pipeline

How a system chooses the final answer, decides when to ask instead, and presents the result

Learning objectives

By the end of this module, you should be able to:

1. Choose between candidate answers using **expected utility**, including uneven costs for different kinds of mistake.
2. Treat the user's explicit requirements as **hard constraints** that come before any score.
3. Raise confidence when **independent derivations agree**.
4. Decide between **answering, hedging and asking** with the **value of information**.
5. Set an answer threshold from a **risk–coverage** trade-off.
6. Match the **presentation format** to the intent, and round numbers to their real precision.
7. Trace one question **end to end** through all six modules.

6.1 The problem: many candidates, one answer

Module 5 produced a ranked list of candidate plans $\pi_1, \ldots, \pi_m$, each with a score. Executing them gives candidate answers $A_1, \ldots, A_m$. Four decisions remain:

Decision	Question	Section
Selection	Which candidate is best?	6.2–6.4
Action	Answer, answer with stated assumptions, or ask first?	6.5
Abstention	When is it better not to answer at all?	6.6
Presentation	What form should the answer take?	6.7

The highest-scoring candidate is *usually* the right choice, but not always. A score measures **probability**, while the decision must also weigh **consequences**.

6.2 Expected utility

Definition

Let w range over the possible "states of the world", such as which interpretation the user meant. Each candidate answer A has a **utility** U(A, w): how good it is if the world is w.

```
EU(A) = Σ_w P(w | Q, evidence) · U(A, w)
A*    = argmax_A EU(A)
```

The simplest case: right or wrong

$$EU(A) = P(correct) \cdot G - P(wrong) \cdot L$$

- **G** is the gain from a correct answer.
- **L** is the loss from a wrong answer that someone acts on.

In business settings, **L is usually larger than G**. A confident wrong number in a board deck costs more than a correct number gains.

Worked example

G = 1 and L = 2.

Candidate	Description	P(correct)	EU
A_1	Certified monthly summary	0.70	0.70(1) - 0.30(2) = **+0.10**
A_2	Aggregated from raw transactions	0.60	0.60(1) - 0.40(2) = **-0.20**

A_1 is the better choice. But **EU = +0.10 is only slightly positive**: with this loss ratio, even the best candidate is close to the point where answering stops being worthwhile.

Break-even confidence

Answering beats saying nothing (EU = 0) when:

$$P \cdot G - (1 - P) \cdot L > 0 \iff P > L / (G + L)$$

Loss ratio L : G	Break-even P
1 : 1	0.50
2 : 1	0.67
4 : 1	0.80
9 : 1	0.90

> **Lesson:** the confidence threshold for answering isn't a fixed number. It follows from **how costly a wrong answer would be**. A high-stakes question needs more certainty.

6.3 Hard constraints come first

Some requirements aren't trade-offs: they're **conditions** that any acceptable answer must meet. Every filter, threshold and grouping the user stated explicitly is a hard constraint κ.

Feasible set, then optimize

```
F = { A : A ⊨ κ  for every explicit constraint κ in Q }
A* = argmax_{A ∈ F} EU(A)
```

This is a **lexicographic** decision: satisfy the constraints first, and compare scores only *within* the feasible set. A higher score never buys back a violated constraint.

Worked example

Question: *"Q3 revenue in EMEA, excluding returns."* Three constraints: κ_1 = Q3, κ_2 = EMEA, κ_3 = excludes returns.

Candidate	Score	κ_1	κ_2	κ_3	Feasible?
A_1 Gross revenue summary	**0.85**	✓	✓	✗ (includes returns)	**No**
A_2 Net revenue summary	0.72	✓	✓	✓	Yes
A_3 Transactions, returns filtered	0.65	✓	✓	✓	Yes

A_2 is chosen. A_1 has the highest score but is removed, however convenient it looks.

When nothing is feasible

```
F = ∅   ⇒   do not relax constraints silently
        ⇒   report which constraint cannot be met, and why
```

For example, if no source tracks returns separately, say so. Don't quietly return gross revenue.

> **Why so strict?** A user who stated a filter will assume it was applied. Silently dropping it gives an answer that **looks** right and **is** wrong, which is the most damaging kind of error.

6.4 Agreement between independent derivations

If two **independent** plans reach the **same** answer, that agreement is strong evidence. Two wrong methods rarely produce the same wrong number.

Bayesian update

Let H = "answer A_1 is correct".

```
P(H | agree) = P(agree | H) · P(H) / [ P(agree | H) · P(H) + P(agree | ¬H) · P(¬H) ]
```

Worked example

- Prior: P(A_1 correct) = 0.70
- A_2, derived independently, is correct with probability 0.60. If A_2 is wrong, it matches A_1 only by coincidence, with probability 0.05.
- If A_1 is wrong, A_2 matches it only by coincidence, with probability 0.05.

```
P(agree | H)  = 0.60 + 0.40 × 0.05 = 0.62
P(agree | ¬H) = 0.05

P(H | agree) = 0.70 × 0.62 / (0.70 × 0.62 + 0.30 × 0.05)
             = 0.434 / (0.434 + 0.015)
             = 0.434 / 0.449 ≈ 0.97
```

Agreement raises confidence from **70% to about 97%**.

Disagreement is also informative

If the two results differ, confidence in A_1 **falls**, and the size of the gap often points to the cause:

Pattern of gap	Likely cause
Exact multiple (2×, 3×)	Fan-out join (Module 3)
Small, steady % gap	Different definition (gross vs. net) or exclusion rule
One period differs	Time-window or partial-period mismatch (Module 4)
Opposite sign	Refunds or credits treated differently

Majority vote over several plans

With n plans, choose the answer most of them agree on:

$$A^* = \operatorname{argmax}_a \sum_j w_j \cdot \mathbb{1}[A_j = a] \qquad (w_j = \text{plan confidence})$$

This is called **self-consistency**. It is only valuable when the plans are **genuinely independent**: different sources or different paths. Five runs of the same plan agreeing proves nothing.

6.5 Answer, hedge or ask? The value of information

When interpretations compete, there are three actions:

Action	What happens
Answer	Commit to the most likely interpretation silently
Hedge	Answer, stating the assumption so the user can correct it
Ask	Clarify first, then answer correctly, at a cost c (time, friction)

Value of perfect information

```
EU(answer)        = max_a Σ_w P(w) · U(a, w)              ← best choice under uncertainty
EU(perfect info)  = Σ_w P(w) · max_a U(a, w)              ← best choice if w were known
VOI               = EU(perfect info) − EU(answer)

Ask  ⟺  VOI > c
```

VOI is **never negative**: knowing more can't make the best decision worse.

Worked example: two interpretations

There are two interpretations, w_1 with P = 0.6 and w_2 with P = 0.4. Utilities: correct = +1, silently wrong = −2, wrong but assumption stated = −0.5. A stated assumption lets the user catch the error quickly, so the loss is smaller.

```
Answer (w₁, silently):  0.6(1) + 0.4(−2)   = 0.6 − 0.8 = −0.20
Hedge  (w₁, stated):    0.6(1) + 0.4(−0.5) = 0.6 − 0.2 = +0.40
Ask:                    1 − c
VOI (vs. silent answer) = 1 − (−0.20) = 1.20
```

Cost of asking c	EU(ask)	Best action
0.3	0.70	**Ask**
0.6	0.40	Tie between ask and hedge
0.8	0.20	**Hedge**

Even a very cheap question doesn't make a *silent* answer worthwhile here. Silent answering is dominated by hedging at every cost.

Same setup, more confidence

With $P(w_1) = 0.9$:

```
Answer: 0.9 − 0.1(2)   = 0.70
Hedge:  0.9 − 0.1(0.5) = 0.85
Ask:    1 − c          = 0.70 at c = 0.3
```

Hedge wins: interruptions aren't worth it when one interpretation is strongly favoured, but the assumption is still worth stating.

Decision summary

```
high confidence, low stakes          →  answer, briefly noting any assumption
moderate confidence or high stakes   →  hedge: answer plus stated assumptions and
alternatives
low confidence and high stakes       →  ask
```

6.6 Abstention: the risk–coverage trade-off

Across *many* questions, an answer threshold τ trades **coverage** against **risk**:

```
Coverage(τ) = fraction of questions answered (Conf ≥ τ)
Risk(τ)     = error rate among the answered questions
```

Worked example: 100 questions

Threshold τ	Answered	Wrong	Coverage	Risk
0.0	100	20	100%	20.0%
0.5	80	10	80%	12.5%
0.7	60	3	60%	5.0%
0.9	30	1	30%	3.3%

Goal: keep risk ≤ 5%. The lowest threshold that meets the target is **τ = 0.7**, which answers 60% of questions.

```
τ* = min { τ : Risk(τ) ≤ target }        ← maximize coverage subject to a risk budget
```

Why calibration matters here

This curve only works if the confidence is **calibrated** (Module 4, section 4.10). If "0.8" really means "60% correct", any threshold gives far more errors than expected.

Abstaining well

"I don't know" isn't the only alternative to an answer. Partial answers are often better:

Situation	Better than a flat refusal
One constraint can't be met	Answer the rest, and explain the gap
Two interpretations are equally likely	Show both, clearly labelled
Data is incomplete for the latest period	Answer for complete periods, and flag the partial one

6.7 Presenting the answer

The answer's form should match the **intent** from Module 4:

Intent	Natural format
lookup / count	One headline number
aggregate	Table or bar chart by group
trend	Line chart over time

compare	Paired values with the difference and % change
rank	Ordered bar chart or table
distribution	Histogram or shares of a whole
explain	Waterfall: start → contributions → end

Answer first

Readers value the beginning of an answer most. Order the content by value: **direct answer** → **key evidence** → **assumptions** → **detail**.

Round to the real precision

Don't report more digits than the uncertainty supports. With relative uncertainty ε:

$$\text{meaningful significant figures} \approx \lfloor -\log_{10} \varepsilon \rfloor + 1$$

ε	$-\log_{10} \varepsilon$	Significant figures	$1,047,382 reported as
5%	1.3	3	$1.05M
0.5%	2.3	4	$1.047M

> Reporting "$1,047,382" when the answer is only good to ±5% implies a precision that isn't there.

Make assumptions visible

Every latent variable inferred in Modules 3–4 (time window, calendar, entity choice, default exclusions) whose confidence was below about 0.9 should appear as a **stated assumption**. That is what makes the hedge in 6.5 work: it turns a -2 into a -0.5.

6.8 The full pipeline, end to end

Question: *"Why did EMEA revenue drop last month compared to the month before, excluding test accounts?"***Today:** 2026-10-21. All figures are hypothetical.

Module 1: Relevance

- C(Q) = {Revenue, Region, Month, TestAccount, plus drivers for "why"}
- Hybrid relevance × prior favours a **certified monthly revenue summary** (P ≈ 0.70).
- "Why" needs a breakdown by segment, so set cover adds **transaction-level** and **customer** sources.

Module 2: Concept mapping

Term	Concept
"revenue"	Revenue (net, the curated default)

"EMEA"	Region value (region context, from the noisy-channel prior)
"test accounts"	TestAccount flag
"last month" / "month before"	Time periods (resolved in Module 4)

Module 3: Implicit relationships

- Steiner tree: Transactions → Customers → Regions, and Transactions → Products, at a total cost of about 0.25, so P ≈ e^(−0.25) ≈ 0.78.
- Revenue flows only **many-to-one**, so there is no fan-out risk.
- The test-account exclusion is **explicit**, so P = 1 (a hard constraint).

Module 4: Intent and entities

- Multiple intents: **explain** (0.85) and **compare** (0.80) → explain template.
- Operator: **SUM**.
- Time: last month = **[2026-09-01, 2026-10-01)**; month before = **[2026-08-01, 2026-09-01)**. Both are complete calendar months, so there's no partial-period trap.

Module 5: Matching, checks and ranking

- Hungarian matching assigns the revenue, region, date and test-flag fields.
- "EMEA" is an **exact** match in the region field.
- Combined stage confidence:

```
0.85 (intent) × 0.70 (source) × 0.90 (joins) × 0.95 (entities) × 0.90 (fields) ≈ 0.46
```

Module 6: Selection

Result (hypothetical): August $5.00M → September $4.40M, so **ΔRevenue = −$0.60M (−12%)**.

Decomposition by country:

Segment	Δ	Share of the change
Germany	−$0.45M	75.0%
France	−$0.20M	33.3%
UK	+$0.05M	−8.3%
Other	$0.00M	0.0%
Total	**−$0.60M**	**100%**

Shares can be above 100% or negative when some segments offset others. Here, the UK's growth partly cancelled out the declines.

Consistency with an independent reference: the certified summary reports September at $4.42M.

```
relative gap = |4.40 − 4.42| / 4.42 ≈ 0.45%
Consistency  = exp(−2 × 0.0045) ≈ 0.991
```

Agreement update (6.4): P(agree | correct) = 0.95 and P(agree | wrong) = 0.10.

```
P(correct | agree) = 0.46 × 0.95 / (0.46 × 0.95 + 0.54 × 0.10)
                   = 0.437 / (0.437 + 0.054) ≈ 0.89
```

Hard constraints: EMEA ✓, both months ✓, test accounts excluded ✓, so the answer is **feasible**.

Action (6.5): confidence of 0.89 → **hedge**: answer directly, stating the calendar-month and net-revenue assumptions.

Presentation (6.7): an explain intent → **waterfall chart**, answer first:

> *EMEA revenue fell 12% (-$0.6M) from August to September, driven mainly by Germany (75% of the decline) and France (33%), partly offset by UK growth. Assumes calendar months and net revenue; test accounts excluded.*

How confidence evolved

After	Confidence	Why it changed
Source selection	0.70	One plausible certified source
All five stages	0.46	Uncertainty compounds (Module 5)
Independent agreement	**0.89**	Two derivations matched (6.4)

6.9 Common pitfalls

Pitfall	Symptom	Fix in the math
Picking the top score blindly	Confident answer that ignores the cost of being wrong	Expected utility with L > G
Trading constraints for score	User's filter silently dropped	Feasible set first, then optimize
Relaxing constraints when F = ∅	"Close enough" answers that mislead	Report the infeasible constraint
Counting dependent agreement	Five reruns "confirm" one error	Only independent plans count
Always asking	User friction on easy questions	Ask only when VOI > c
Never asking	Silent misinterpretations	Hedge or ask when VOI is high

Fixed threshold, uncalibrated scores	Risk far above the target	Calibrate, then set τ from risk–coverage
False precision	"$1,047,382" from a ±5% estimate	Round to the supported significant figures
Hidden assumptions	User can't spot the misreading	State every low-confidence inferred variable

Check your understanding

1. **Compute:** P(correct) = 0.8, G = 1 and L = 3. What is EU(answer)? What break-even confidence does this loss ratio imply?
2. **Compute:** two equally likely interpretations (0.5 each), with utility +1 if correct and -1 if wrong. What are EU(answer), EU(perfect info) and VOI? When should the system ask?
3. **Reason:** the top-scoring candidate violates one of the user's stated filters. What should happen?
4. **Compute:** prior P(correct) = 0.5, P(agree | correct) = 0.9 and P(agree | wrong) = 0.1. What is P(correct | agree)?
5. **Compute:** at some threshold, 50 of 100 questions are answered and 2 of those are wrong. What are the coverage and the risk?
6. **Reason:** why can hedging beat both silent answering and asking?

<details>
<summary>Answers</summary>

1. EU = 0.8(1) - 0.2(3) = 0.8 - 0.6 = **+0.20**. Break-even P = 3 / (1 + 3) = **0.75**.
2. EU(answer) = 0.5(1) + 0.5(-1) = **0**. EU(perfect info) = **1**. VOI = **1**. Ask whenever the cost of asking c < 1.
3. It is **removed from the feasible set**, whatever its score. Choose the best feasible candidate. If none exists, report which constraint can't be met.
4. 0.5 × 0.9 / (0.5 × 0.9 + 0.5 × 0.1) = 0.45 / 0.50 = **0.90**.
5. Coverage = **50%**. Risk = 2 / 50 = **4%**.
6. Hedging keeps most of the benefit of answering immediately (no cost c) while shrinking the loss if wrong, because a stated assumption lets the user correct it quickly. When confidence is moderate and asking is costly, that combination has the highest expected utility.

</details>

Key takeaways

- **Expected utility**, not raw score, picks the answer, and the threshold for answering follows from the **cost of a wrong answer**.
- The user's explicit requirements are **hard constraints**. Filter first, optimize second, and never relax them silently.
- **Independent agreement** sharply raises confidence, and the pattern of a **disagreement** points to its cause.
- **Answer, hedge or ask** is a value-of-information decision. Hedging is often the best middle ground.
- Thresholds come from a **risk–coverage** target and only work with **calibrated** confidence.
- **Presentation** follows intent: lead with the answer, round to the real precision, and state assumptions.

Course recap

Module	Core question	Key mathematical tools

1. Relevance	Which sources matter?	Cosine similarity, BM25, concept coverage, Bayesian priors, set cover
2. Concept mapping	Do these words mean the same thing?	Equivalence classes, edit distance, softmax and entropy, noisy channel, PMI, ontology similarity
3. Implicit relationships	How do things connect, and what went unstated?	Schema graphs, -log P costs, shortest paths, Steiner trees, conditional entropy, latent variables
4. Intent and entities	What shape of answer, about which things?	Naive Bayes, templates, aggregation rules, BIO tagging, joint assignment, calibration
5. Matching, confidence and ranking	Which fields? How sure? What order?	Hungarian algorithm, stage confidence, robust z-scores, learning to rank, MMR, nDCG
6. Answer selection	Which answer, and how is it delivered?	Expected utility, hard constraints, Bayesian agreement, value of information, risk–coverage

The whole pipeline in one line:

$$A^* = \operatorname{argmax}_{A \in F(Q)} \sum_w P(w \mid Q, D, \mathcal{O}, \text{evidence}) \cdot U(A, w)$$

In words: among the answers that respect everything the user asked for, choose the one with the best expected outcome, given everything learned about what the question means and how the data fits together. If no answer clears the bar, ask.